

Malware Analysis Report

AsyncRAT v0.5.8

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*Static & Dynamic Analysis of a
Remote Access Trojan*

Threat Level: MALICIOUS

Tools Used

PEStudio 9.61 | Detect-it-Easy 3.2.0 | Any.run Sandbox
MITRE ATT&CK

1. Executive Summary

A malicious archive was submitted for analysis via MalwareBazaar and found to contain AsyncRAT v0.5.8, a well-known Remote Access Trojan (RAT) capable of remote system control, keylogging, credential theft, and data exfiltration. The sample was identified through static analysis using PEStudio, Detect-it-Easy and dynamic detonation in the any.run interactive sandbox.

The archive was delivered as a password-protected ZIP file, a common evasion technique used to bypass email gateway scanning. Upon extraction via WinRAR, the embedded executable was staged in the system Temp directory and launched. Once executed, AsyncRAT immediately performed victim fingerprinting by querying the Windows Registry for the machine GUID, computer name, and supported system languages — information used to profile the target before establishing command and control (C2) communications.

The sample was configured with two C2 domains (lytuong.net and 123b-mobilee.com) and eight communication ports, including ports 443 and 80 to disguise malicious traffic as legitimate HTTPS and HTTP traffic. Notably, AntiVM and AutoRun were both disabled in this sample's configuration, suggesting a broad phishing campaign rather than a targeted attack, and that persistence was not established during this execution.

The threat was classified as Malicious by the any.run sandbox with a score of 100/100, and AsyncRAT was detected via YARA rules during analysis.

Malware Family	AsyncRAT v0.5.8
Threat Level	MALICIOUS (100/100)
Persistence Observed	No
AntiVM / AntiSandbox	Disabled
C2 Infrastructure	lytuong.net 123b-mobilee.com

2. Sample Details & Indicators of Compromise (IOCs)

The following table provides all indicators of compromise extracted from the sample during static and dynamic analysis. These should be immediately actioned for detection and blocking across the environment.

Field	Value
Filename	e783518120cd5ce9ee99f183ce92b9a94456ba8423b4d3fbb95690107d256385.zip
SHA256 (ZIP)	e783518120cd5ce9ee99f183ce92b9a94456ba8423b4d3fbb95690107d256385
SHA256 (EXE)	25A3D982C7C731D689F205BF83A5B0A303DBC3BBA660E88A114C51E176986A88
File Size	22,854 bytes (packed) / 46,080 bytes (unpacked)
Entropy	7.992 (packed at 99%)
Malware Family	AsyncRAT
Version	0.5.8
Botnet Tag	Default
Mutex	FtTRmAL1FxC0
Install Path	%AppData%
C2 Domain 1	lytuong.net
C2 Domain 2	123b-mobilee.com
C2 Ports	6606, 7707, 8808, 443, 80, 8844, 22, 53
AutoRun	false
AntiVM	false
BSoD	false
Analysis Date	24 May 2026

3. Static Analysis

Static analysis was performed using PEStudio 9.61, a free malware initial assessment tool.

The analysis was conducted without executing the sample, examining file metadata, strings, entropy and hash values.

3.1 File Identification (PEStudio — Indicators)

PEStudio identified 4 indicators of interest, including the file name, SHA256 hash, file size and entropy. The file reported an entropy of 7.992, which is extremely high (maximum is 8.0), strongly indicating the file is packed or encrypted — a common technique to evade antivirus detection and complicate static analysis.

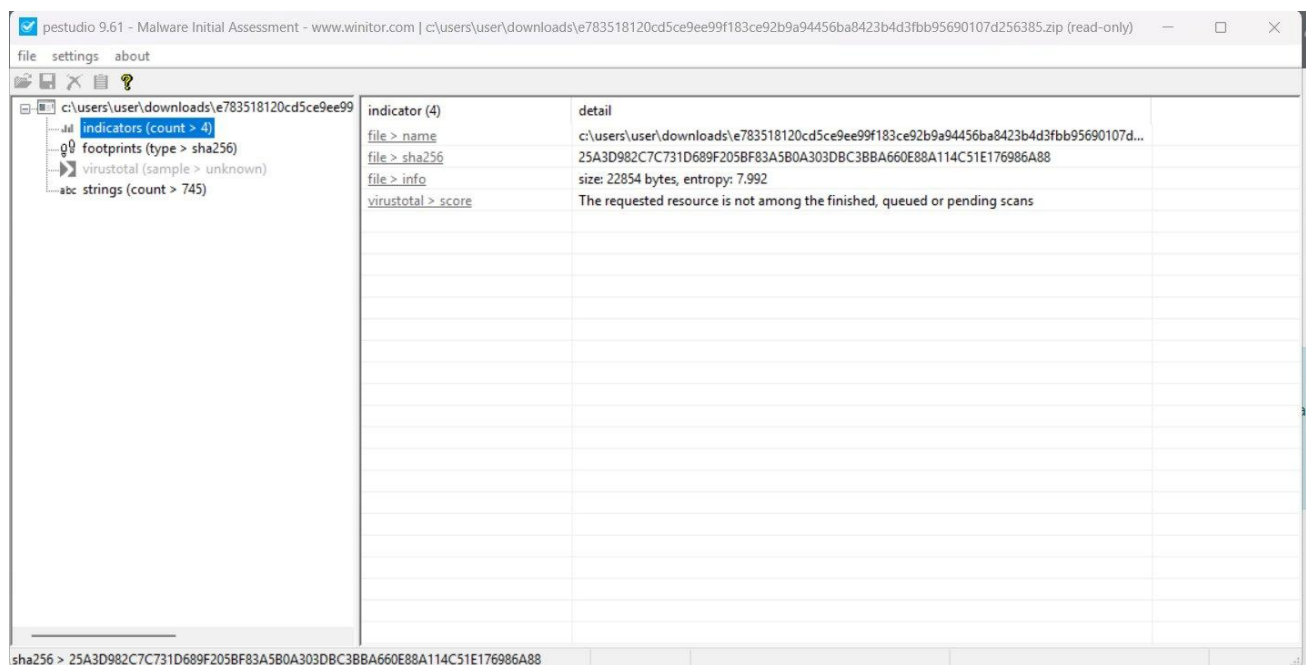


Figure 1: PEStudio indicators panel showing file metadata including SHA256 hash and entropy score

3.2 String Analysis

The strings panel in both PESTudio and Detect-it-Easy revealed 745 strings embedded within the archive. The binary view confirmed the presence of the executable filename as a readable ASCII string, along with numerous obfuscated or garbage strings — a hallmark of packed malware designed to defeat string-based detection signatures.

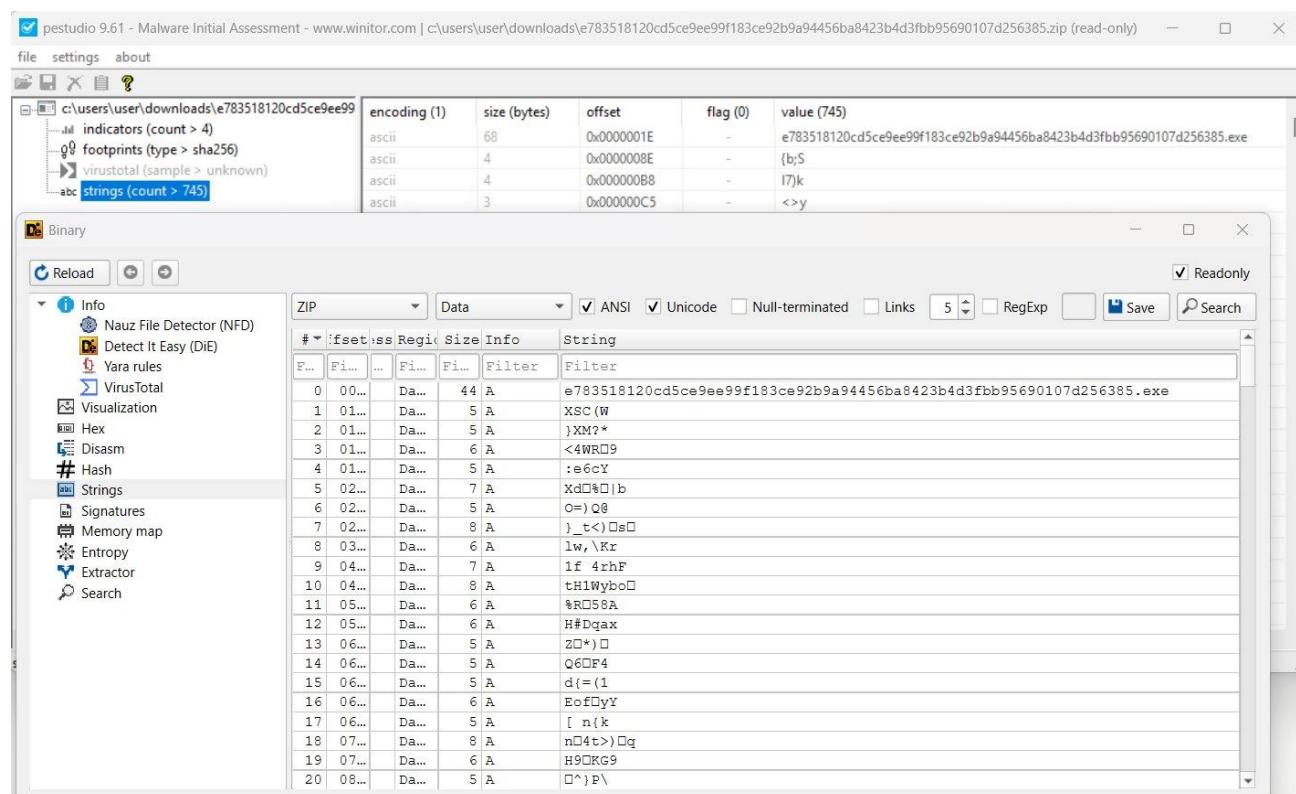


Figure 2: PESTudio and Detect-it-Easy string analysis showing 745 strings — the executable filename is visible as the first readable string

3.3 Entropy Analysis

The entropy diagram in the binary viewer confirmed the sample is packed at 99%, with an overall entropy of 7.99158 across the entire data section. The flat entropy curve visible in the diagram is characteristic of encrypted or compressed payloads, as randomness is uniformly distributed throughout the file rather than varying by section as seen in legitimate executables.

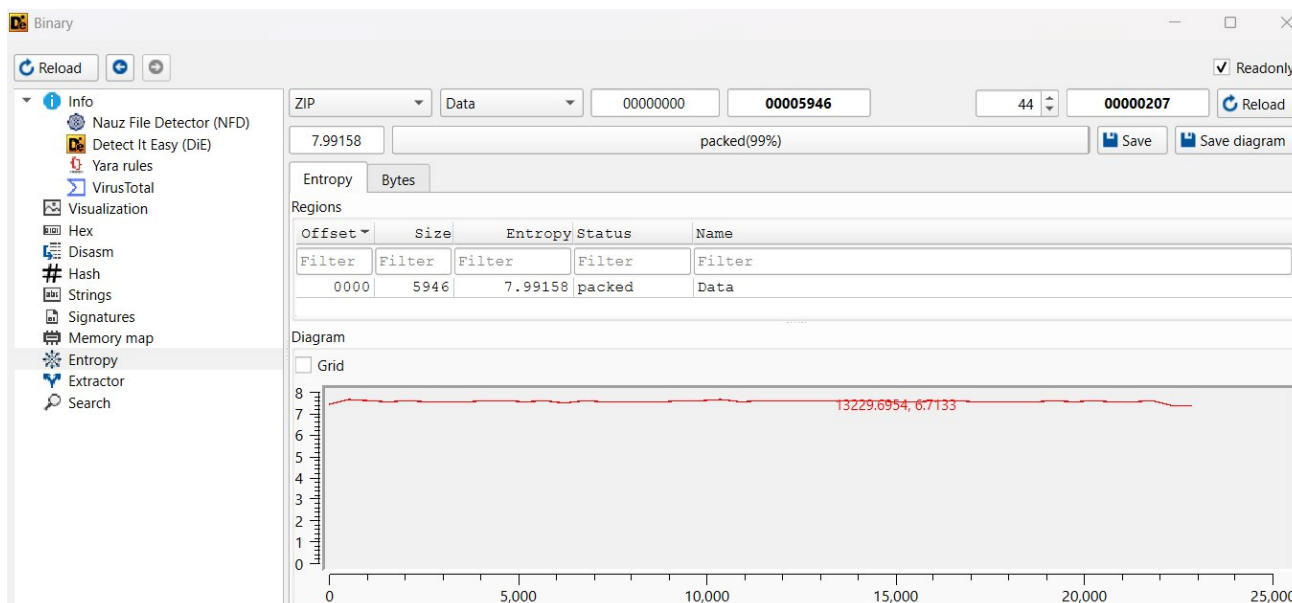


Figure 3: Entropy analysis showing 7.99 entropy and packed (99%) status — consistent with an encrypted or compressed payload

4. Dynamic Analysis — Sandbox Execution

Dynamic analysis was performed using the any.run interactive sandbox running Windows 10 64-bit. The sample was detonated in a controlled environment and all behavioural activity was captured including process creation, registry access, network traffic, and file system changes.

4.1 Execution Flow

The following steps describe the complete execution chain observed during sandbox analysis:

- Step 1 — The user opened the malicious ZIP archive using WinRAR.exe from C:\Program Files\WinRAR\
- Step 2 — WinRAR extracted the embedded executable to C:\Users\admin\AppData\Local\Temp\Rar\$EXb7664.30156\
- Step 3 — The extracted executable was launched (PID 6864), spawning a second instance (PID 4236)
- Step 4 — The malware was identified as AsyncRAT by YARA rule detection with a malicious score of 100/100
- Step 5 — AsyncRAT immediately queried the Windows Registry to fingerprint the victim system
- Step 6 — The malware attempted to establish C2 communications with configured domains

4.2 Archive Extraction (any.run — WinRAR)

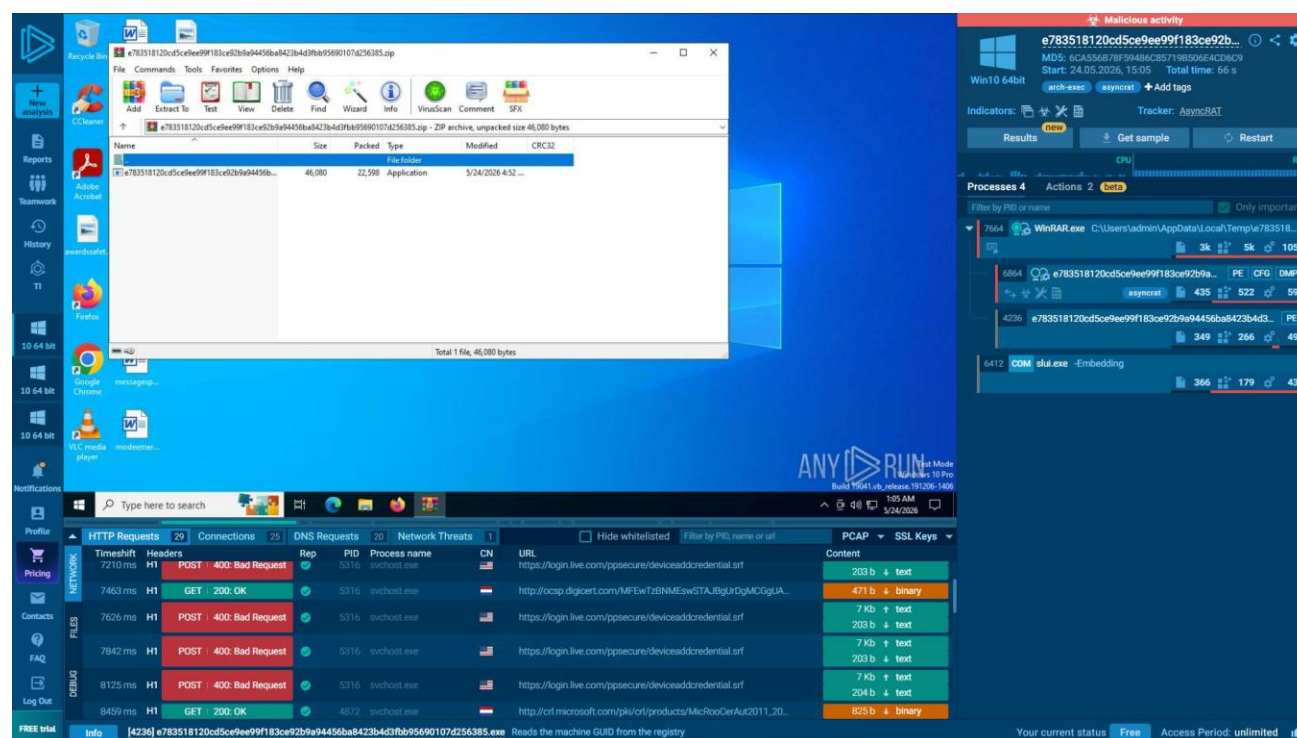


Figure 4: any.run sandbox showing the malicious ZIP opened in WinRAR with AsyncRAT tracker confirmed (top right). Process tree shows WinRAR.exe spawning two instances of the payload executable.

4.3 Extraction in Progress

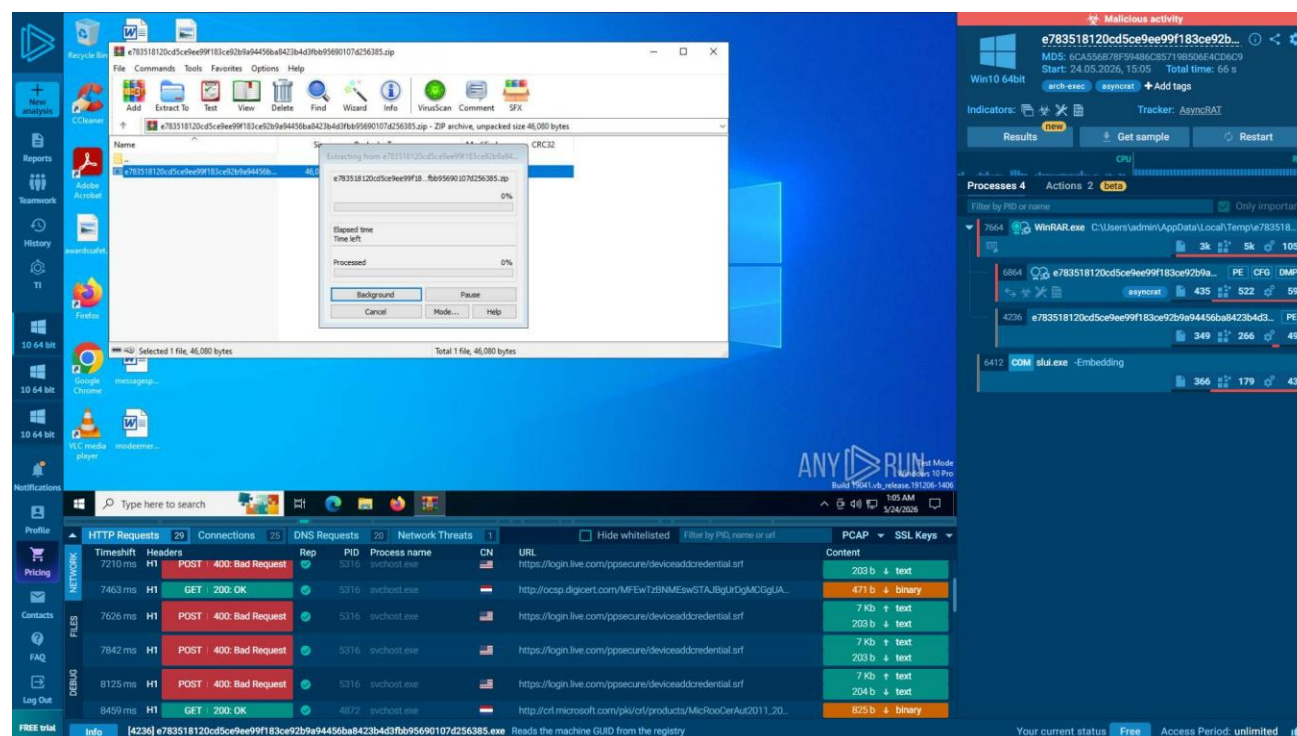


Figure 5: WinRAR extraction dialog captured mid-execution — the payload is being written to the Temp directory

4.4 Process Tree — AsyncRAT Detected

The process graph from any.run clearly shows the parent-child execution chain. WinRAR.exe spawns the malicious executable, which is immediately flagged with the ASYNCRAT label and a biohazard icon. The process detail panel confirms YARA detection with a danger score of 100/100.

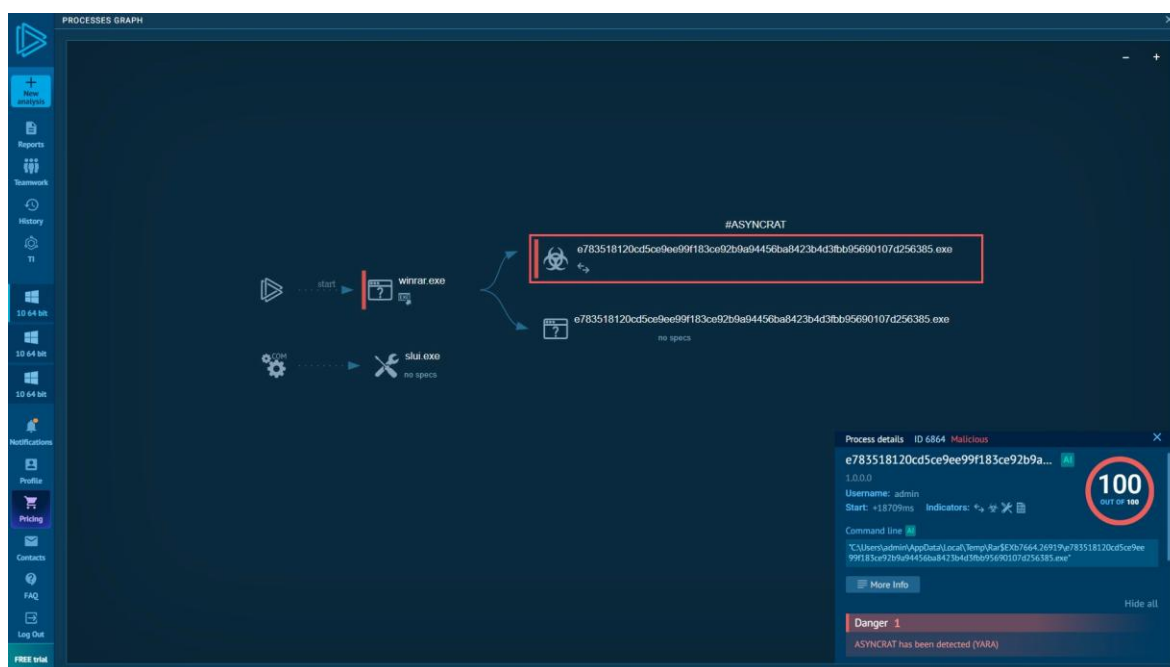


Figure 6: any.run process graph — WinRAR.exe spawns AsyncRAT payload (PID 6864, score 100/100). YARA rule 'ASYNCRAT has been detected' triggered.

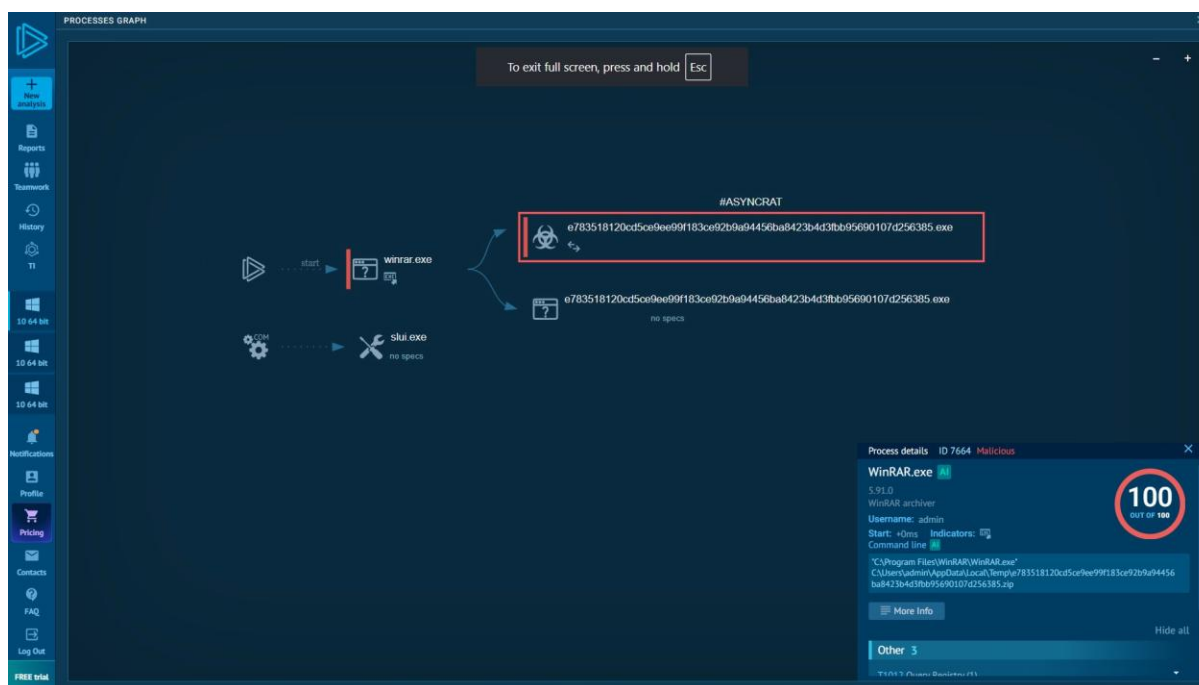


Figure 7: Process details showing WinRAR.exe (PID 7664) command line — the full path to the malicious ZIP is visible confirming the execution chain

4.5 Malware Configuration (C2 & Options)

any.run successfully extracted the full AsyncRAT configuration from memory during detonation. This revealed the complete C2 infrastructure including domains, ports, version, mutex, and operational flags.

The screenshot shows the 'Malware configuration' window in any.run. The window title is 'Malware configuration' and it contains the text 'Here are the details of the configuration'. Below this, there is a tab labeled 'AsyncRat (1)' with options to 'Hide all', 'Copy selected (0)', and 'Download JSON'. The main content area is divided into two panes. The left pane shows a table of configuration items, and the right pane shows the corresponding JSON configuration file.

No data	
C2 (2)	lytuong.net
	123b-mobilee.com
Ports (8)	6606
	7707
	8808
	443
	80
	8844
	22
	53
Version	0.5.8
Botnet	Default
Options	
AutoRun	false
Mutex	FTTRmAL1FxCo
InstallFolder	%AppData%
BSoD	false
AntiVM	false
Certificates	

```
{
  "C2": [
    "lytuong.net",
    "123b-mobilee.com"
  ],
  "Ports": [
    6606,
    7707,
    8808,
    443,
    80,
    8844,
    22,
    53
  ],
  "Version": "0.5.8",
  "Botnet": "Default",
  "Options": {
    "AutoRun": false,
    "Mutex": "FTTRmAL1FxCo",
    "InstallFolder": "%AppData%",
    "BSoD": false
  },
  "AntiVM": false
}
```

Figure 8: Extracted AsyncRAT malware configuration showing C2 domains (lytuong.net, 123b-mobilee.com), 8 communication ports, version 0.5.8, mutex, and configuration options including AntiVM: false

5. MITRE ATT&CK Mapping

The behaviours observed during analysis were mapped to the MITRE ATT&CK framework v19 (Enterprise). Five techniques were identified across the Execution, Discovery, and Command & Control tactic categories.

Technique ID	Name	Tactic	Observed Behaviour
T1204.002	Malicious File	Execution	User opened a ZIP archive containing a disguised executable, extracted via WinRAR
T1012	Query Registry	Discovery	Malware read machine GUID, computer name, and supported languages from the Windows Registry
T1082	System Information Discovery	Discovery	Collected system information to fingerprint the victim and shape follow-on C2 behaviour
T1071	Application Layer Protocol	C&C	C2 traffic configured across ports 443 and 80 to blend with legitimate web traffic
T1105	Ingress Tool Transfer	C&C	Executable staged from archive into the Temp directory prior to execution

5.1 Discovery Tactic (TA0007)

The screenshot displays the MITRE ATT&CK framework interface. The top navigation bar includes links for Matrices, Tactics, Techniques, Defenses, CTI, Resources, Benefactors, Contribute, and Blog. The main content area is titled "Discovery" and includes a description: "The adversary is trying to figure out your environment." Below this, a table lists techniques related to Discovery. The sidebar on the left shows a hierarchy of tactics, with "Discovery" selected.

ID	Name	Description
T1087	Account Discovery	Adversaries may attempt to get a listing of valid accounts, usernames, or email addresses on a system or within a compromised environment. This information can help adversaries determine which accounts exist, which can aid in follow-on behavior such as brute-forcing, spear-phishing attacks, or account takeovers (e.g., Valid Accounts).
.001	Local Account	Adversaries may attempt to get a listing of local system accounts. This information can help adversaries determine which local accounts exist on a system to aid in follow-on behavior.
.002	Domain Account	Adversaries may attempt to get a listing of domain accounts. This information can help adversaries determine which domain accounts exist to aid in follow-on behavior such as targeting specific accounts which possess particular privileges.
.003	Email Account	Adversaries may attempt to get a listing of email addresses and accounts. Adversaries may try to dump Exchange address lists such as global address lists (GALs).
.004	Cloud Account	Adversaries may attempt to get a listing of cloud accounts. Cloud accounts are those created and configured by an organization for use by users, remote support, services, or for administration of resources within a cloud service provider or SaaS application.

Figure 9: MITRE ATT&CK Discovery tactic page — AsyncRAT performed system discovery to understand the victim environment

5.2 T1012 — Query Registry

MITRE | ATT&CK

Matrices | Tactics | Techniques | Defenses | CTI | Resources | Benefactors | Contribute | Blog | Search Q

ATT&CK v19 has been released! Check out the [blog post](#) for more information.

Home > Techniques > Enterprise > Query Registry

Query Registry

Adversaries may interact with the Windows Registry to gather information about the system, configuration, and installed software.

The Registry contains a significant amount of information about the operating system, configuration, software, and security.^[1] Information can easily be queried using the `Reg` utility, though other means to access the Registry exist. Some of the information may help adversaries to further their operation within a network. Adversaries may use the information from Query Registry during automated discovery to shape follow-on behaviors, including whether or not the adversary fully infects the target and/or attempts specific actions.

ID: T1012
Sub-techniques: No sub-techniques
Tactic: Discovery
Platforms: Windows
Version: 1.3
Created: 31 May 2017
Last Modified: 12 May 2026

[Version Permalink](#)

Procedure Examples

ID	Name	Description
S0045	ADVSTORESHELL	ADVSTORESHELL can enumerate registry keys. ^{[2][3]}
G0050	APT32	APT32's backdoor can query the Windows Registry to gather system information. ^[4]
G0087	APT39	APT39 has used various strains of malware to query the Registry. ^[5]
G0096	APT41	APT41 queried registry values to determine items such as configured RDP ports and network configurations. ^[6]
S0438	Attor	Attor has opened the registry and performed query searches. ^[7]
S0344	Azorult	Azorult can check for installed software on the system under the Registry key <code>Software\Microsoft\Windows\CurrentVersion\Uninstall</code> . ^[8]

Figure 10: MITRE ATT&CK T1012 Query Registry — observed when AsyncRAT read machine GUID and computer name from the Windows Registry

5.3 T1518 — Software Discovery

MITRE | ATT&CK

Matrices | Tactics | Techniques | Defenses | CTI | Resources | Benefactors | Contribute | Blog | Search Q

ATT&CK v19 has been released! Check out the [blog post](#) for more information.

Home > Techniques > Enterprise > Software Discovery

Software Discovery

Sub-techniques (2)

Adversaries may attempt to get a listing of software and software versions that are installed on a system or in a cloud environment. Adversaries may use the information from Software Discovery during automated discovery to shape follow-on behaviors, including whether or not the adversary fully infects the target and/or attempts specific actions.

Such software may be deployed widely across the environment for configuration management or security reasons, such as Software Deployment Tools, and may allow adversaries broad access to infect devices or move laterally.

Adversaries may attempt to enumerate software for a variety of reasons, such as figuring out what security measures are present or if the compromised system has a version of software that is vulnerable to Exploitation for Privilege Escalation.

ID: T1518
Sub-techniques: T1518.001, T1518.002
Tactic: Discovery
Platforms: ESXi, IaaS, Linux, Windows, macOS
Version: 1.5
Created: 16 September 2019
Last Modified: 12 May 2026

[Version Permalink](#)

Procedure Examples

ID	Name	Description
S0534	Bazar	Bazar can query the Registry for installed applications. ^[1]
G0060	BRONZE BUTLER	BRONZE BUTLER has used tools to enumerate software installed on an infected host. ^[2]
S0482	Bundlore	Bundlore has the ability to enumerate what browser is being used as well as version information for Safari. ^[3]
S0674	CharmPower	CharmPower can list the installed applications on a compromised host. ^[4]
S0154	Cobalt Strike	The Cobalt Strike System Profiler can discover applications through the browser and identify the version of Java the target has. ^[5]

Figure 11: MITRE ATT&CK T1518 Software Discovery — malware checked supported languages and system configuration to profile the target

6. Recommendations

6.1 Immediate Actions

- Block lytuong.net and 123b-mobilee.com at DNS, proxy, and firewall level
- Block outbound connections on non-standard ports 6606, 7707, 8808, 8844 from endpoints
- Scan the environment for the mutex FtTRmAL1FxCo to identify any active infections
- Search for the SHA256 hash
e783518120cd5ce9ee99f183ce92b9a94456ba8423b4d3fbb95690107d256385 across endpoint logs
- Quarantine any files matching the IOCs listed in Section 2

6.2 Detection Rules

- Create a SIEM alert: executable spawned by WinRAR.exe from any Temp or AppData path
- Enable Sysmon Event ID 1 (process creation), 12 and 13 (registry access) for monitoring
- Alert on Windows Security Event ID 4688 for processes launched from user-writable directories
- Monitor DNS queries and proxy logs for the identified C2 domains

6.3 Hardening

- Implement AppLocker or WDAC policies to block execution from %AppData%, %Temp%, and Downloads
- Enforce email gateway scanning that rejects or sandboxes password-protected archive attachments
- Auto-submit downloaded archives to a sandbox (e.g., any.run or Windows Defender Sandbox) before allowing execution
- Train users to recognise social engineering lures that deliver executable-in-zip payloads
- Seed canary/decoy files in common staging paths and alert on unexpected access or execution

7. References

MalwareBazaar — Abuse.ch malware repository: <https://bazaar.abuse.ch>

any.run — Interactive sandbox analysis platform: <https://any.run>

PEStudio — Malware initial assessment tool: <https://www.winitor.com>

Detect-it-Easy — File type identification tool: <https://apps.microsoft.com/detail/9nq58d7ghb2x>

MITRE ATT&CK Framework v19 — Enterprise techniques: <https://attack.mitre.org>

T1204.002 — Malicious File: <https://attack.mitre.org/techniques/T1204/002/>

T1012 — Query Registry: <https://attack.mitre.org/techniques/T1012/>

T1082 — System Information Discovery: <https://attack.mitre.org/techniques/T1082/>

T1071 — Application Layer Protocol: <https://attack.mitre.org/techniques/T1071/>

T1105 — Ingress Tool Transfer: <https://attack.mitre.org/techniques/T1105/>

AsyncRAT — Open-source RAT repository (for reference): <https://github.com/NYAN-x-CAT/AsyncRAT-C-Sharp>